

(MATH203) Applied Linear Algebra

Problems

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Problems are arranged chronologically by year and semester, with the midterm preceding the final examination within the same semester, while preserving the original problem order within each source.

Problem 1. (2018 Fall POSTECH Applied Linear Algebra) In this problem, V is a vector space. Determine whether each of the following statements is true or false.

- (1) There exists a linearly independent subset $S \subset V$ that contains the zero vector $\mathbf{0}$.
- (2) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible, then the columns of $\mathbf{A}$ form a basis for $\mathbb{R}^n$.
- (3) Every projection matrix is invertible.
- (4) $\text{Null}(\mathbf{A})^\perp = \text{Col}(\mathbf{A})$.
- (5) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V , and let $T : V \rightarrow W$ be a linear transformation. If

$$T(\mathbf{v}_i) = \mathbf{w}_i, \quad 1 \leq i \leq n,$$

for given vectors $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$, then T is uniquely determined.

Problem 2. (2018 Fall POSTECH Applied Linear Algebra) For each statement, prove it or disprove it by providing a counterexample.

- (1) Consider a linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{b} \in \mathbb{R}^m.$$

If there are more variables than equations, $m < n$, then the system always has a solution.

- (2) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfies

$$\mathbf{A}^2 = \mathbf{A} \quad \text{and} \quad \text{rk}(\mathbf{A}) = n,$$

then $\mathbf{A} = \mathbf{I}_n$.

- (3) The null space of $\mathbf{A}$ is equal to the null space of $\mathbf{A}^\top \mathbf{A}$.

Problem 3. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

- (a) Find $\mathbf{A}^{-1}$.
- (b) Compute $(\mathbf{A}\mathbf{A}^\top)^{-1}$.

Problem 4. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

- (a) Find a right inverse $\mathbf{B}$ such that

$$\mathbf{A}\mathbf{B} = \mathbf{I}_2.$$
- (b) Using the matrix $\mathbf{B}$ obtained in part (a), compute $(\mathbf{B}\mathbf{A})^{100}$.

Problem 5. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 4 & 4 \\ 4 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Find the orthogonal projection $\mathbf{b}_c$ of $\mathbf{b}$ onto $\text{Col}(\mathbf{A})$.

Problem 6. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{pmatrix}.$$

- Find an LU decomposition $\mathbf{A} = \mathbf{L}\mathbf{U}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.
- Find a basis for $\text{Col}(\mathbf{B})$.
- Find $\text{Null}(\mathbf{B})$.
- Write the complete solution in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ for

$$\mathbf{C} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix},$$

where $\mathbf{x}_p$ is a particular solution and $\mathbf{x}_n \in \text{Null}(\mathbf{C})$.

Problem 7. (2018 Fall POSTECH Applied Linear Algebra) Let $V = M_2(\mathbb{R})$, the vector space of 2×2 real matrices with the usual addition and scalar multiplication.

- Show that

$$\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

form a basis for V .

- Find the matrix representation of the linear transformation

$$T : V \rightarrow V, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a+d & c \\ b & a-d \end{pmatrix},$$

with respect to the ordered basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

Problem 8. (2018 Fall POSTECH Applied Linear Algebra) Let

$$P_k = \left\{ a_k t^k + a_{k-1} t^{k-1} + \cdots + a_1 t + a_0 \mid a_0, \dots, a_k \in \mathbb{R} \right\},$$

with the usual polynomial addition and scalar multiplication. Use the ordered basis

$$\{1, t, t^2, \dots, t^k\}$$

for P_k .

- Find the matrix representation $\mathbf{A}$ of the differentiation map

$$D : P_2 \rightarrow P_1, \quad D(f) = \frac{df}{dt}.$$

- Find the matrix representation $\mathbf{B}$ of the integration map

$$T : P_1 \rightarrow P_2, \quad (Tf)(t) = \int_0^t f(s) \, ds.$$

Problem 9. (2018 Fall POSTECH Applied Linear Algebra) Determine whether each of the following statements is true or false.

- (1) $\mathbf{A}$ and $\mathbf{A}^T \mathbf{A}$ have the same rank for every rectangular matrix $\mathbf{A}$.
- (2) The only possible eigenvalues of a symmetric orthogonal matrix are 1 and -1 .
- (3) For an $n \times n$ matrix $\mathbf{A}$ and a scalar $k \in \mathbb{R}$,

$$\det(k\mathbf{A}) = k \det(\mathbf{A}).$$

- (4) For an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

- (5) If there is a basis consisting of eigenvectors of a matrix, then the matrix is diagonalizable.
- (6) Every diagonalizable matrix has an orthonormal basis consisting of eigenvectors.

Problem 10. (2018 Fall POSTECH Applied Linear Algebra) Compute the determinants of the following matrices.

- (a) Use cofactor expansion to find the determinant of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -1 \\ -1 & 1 & 3 \end{pmatrix}.$$

- (b) Transform the matrix into upper triangular form using elementary row operations, and compute the determinant of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 2 & 4 & 0 \end{pmatrix}.$$

Problem 11. (2018 Fall POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be real square matrices. Prove that if $\mathbf{A}$ and $\mathbf{B}$ are orthogonally similar, then they have the same singular values.

Problem 12. (2018 Fall POSTECH Applied Linear Algebra) Write down as many pairwise nonsimilar 4×4 matrices as possible whose only eigenvalue is 2.

Problem 13. (2018 Fall POSTECH Applied Linear Algebra) Consider $\mathbb{R}^4$ with the standard inner product and let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \subset \mathbb{R}^4,$$

where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 3 \\ 2 \\ -3 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$$

- (a) Find a subset of S that is a basis for $\text{Span}(S)$ and includes $\{\mathbf{v}_1, \mathbf{v}_2\}$.
- (b) Apply the Gram–Schmidt process to the subset found in part (a) and obtain an orthonormal basis for $\text{Span}(S)$.

Problem 14. (2018 Fall POSTECH Applied Linear Algebra) Identify and sketch the curve

$$x^2 + xy + y^2 = 1.$$

Justify your identification with an explicit equation. Indicate all x -intercepts and y -intercepts, and label all coordinate axes.

Problem 15. (2018 Fall POSTECH Applied Linear Algebra) Solve the initial value problem

$$\mathbf{u}'(t) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \mathbf{u}(t), \quad \mathbf{u}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Problem 16. (2018 Fall POSTECH Applied Linear Algebra) Use a similarity transformation to find the matrix $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ representing reflection across the plane

$$x + y + z = 0.$$

Problem 17. (2018 Fall POSTECH Applied Linear Algebra) Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 3 \\ 0 & 0 & 5 \end{pmatrix}.$$

Problem 18. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}.$$

- (a) Find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is diagonal.
- (b) Find the characteristic polynomial of $\mathbf{A}$.
- (c) Show that $\mathbf{A}^{-1}$ can be expressed as

$$a\mathbf{A}^2 + b\mathbf{A} + c\mathbf{I}$$

for some $a, b, c \in \mathbb{R}$, and find a, b, c .

- (d) Find $e^{\mathbf{A}}$.

Problem 19. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find all eigenvalues of $\mathbf{A}$.
- (b) Find all eigenvectors of $\mathbf{A}$.
- (c) Determine whether $\mathbf{A}$ is diagonalizable. Explain why.
- (d) Find the Jordan form of $\mathbf{A}$.
- (e) Find a Jordan basis of $\mathbf{A}$.
- (f) Find $\det(e^{\mathbf{A}})$.
- (g) Calculate $e^{\mathbf{A}}$.

(h) Solve

$$\mathbf{y}' = \mathbf{A}\mathbf{y}, \quad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}.$$

Problem 20. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{D} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix},$$

where $\mathbf{A}$ and $\mathbf{C}$ are square matrices and $\mathbf{0}$ is a zero matrix of compatible size. Prove that

$$\det(\mathbf{D}) = \det(\mathbf{A}) \det(\mathbf{C}).$$

Problem 21. (2019 Spring POSTECH Applied Linear Algebra) Let (g_k) be a sequence satisfying

$$g_{k+2} = g_{k+1} - g_k, \quad g_1 = 1, \quad g_2 = 2.$$

(a) Find g_{300} .

(b) Determine whether the sequence (g_k) converges. In either case, explain why.

Problem 22. (2019 Spring POSTECH Applied Linear Algebra) Calculate

$$\det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Problem 23. (2019 Spring POSTECH Applied Linear Algebra) Let

$$f(x) = \det \begin{pmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & x & x^2 & x^3 \end{pmatrix}.$$

Find all zeros of $f(x)$ and verify your answer.

Problem 24. (2019 Spring POSTECH Applied Linear Algebra) Let $\mathbf{M} \in \mathbb{R}^{n \times n}$ be invertible, and let $\mathbf{v}_1, \dots, \mathbf{v}_l \in \mathbb{R}^n$ be linearly independent. Prove that

$$\mathbf{M}\mathbf{v}_1, \dots, \mathbf{M}\mathbf{v}_l$$

are also linearly independent.

Problem 25. (2019 Spring POSTECH Applied Linear Algebra) Prove that the two matrices

$$\begin{pmatrix} \mathbf{A}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mathbf{A}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_1 \end{pmatrix}$$

are similar, where $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are square matrices of sizes m, n, l , respectively.

Problem 26. (2019 Spring **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1+i \\ 0 & 1 & 1-i \\ 0 & 0 & i \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{A}$.
- (b) Find the eigenvectors of $\mathbf{A}$.
- (c) Find a matrix $\mathbf{S}$ such that $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ is diagonal.

Problem 27. (2019 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ be a real symmetric matrix, and let λ be an eigenvalue of $\mathbf{A}$ with algebraic multiplicity m . Show that

$$\dim \text{Null}(\mathbf{A} - \lambda\mathbf{I}) = m.$$

Problem 28. (2019 Spring **POSTECH** Applied Linear Algebra)

- (1) State and prove Schur's Lemma.
- (2) State and prove the complex spectral theorem.

Problem 29. (2019 Fall **POSTECH** Applied Linear Algebra) Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} 1, & i - j = 1 \text{ and } i < n, \\ 0, & \text{otherwise.} \end{cases}$$

Find the sum of all entries of

$$\sum_{k=1}^{2n} \mathbf{A}^k.$$

Problem 30. (2019 Fall **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} a & 0 & b \\ a & a & 4 \\ 0 & a & 2 \end{pmatrix}.$$

For what values of a and b does the system

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 2 \\ 4 \\ b \end{pmatrix}$$

have

- (1) a unique solution,
- (2) a one-parameter family of solutions,
- (3) a two-parameter family of solutions,
- (4) no solution?

Problem 31. (2019 Fall **POSTECH** Applied Linear Algebra) Find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & -3 & 0 \\ 1 & -2 & 0 & 4 \\ -1 & 2 & 0 & 4 \\ 1 & 2 & 3 & 0 \end{pmatrix}.$$

Problem 32. (2019 Fall POSTECH Applied Linear Algebra) Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be two bases of a vector space V . If

$$\mathbf{e}_i = \sum_{j=1}^n a_{ij} \mathbf{f}_j, \quad i = 1, \dots, n,$$

show that the matrix $\mathbf{A} = (a_{ij})$ is invertible.

Problem 33. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 5 & 1 & 1 \\ 1 & 1 & 3 & 0 & -1 \\ 2 & 2 & 12 & 1 & 0 \end{pmatrix}.$$

- (a) Find a basis for $\text{Row}(\mathbf{A})$ and a basis for $\text{Col}(\mathbf{A})$.
 (b) Consider the restriction

$$\mathbf{A}|_{\text{Row}(\mathbf{A})} : \text{Row}(\mathbf{A}) \rightarrow \text{Col}(\mathbf{A}).$$

Find its matrix representation with respect to the bases obtained in part (a), and verify that this matrix is invertible.

Problem 34. (2019 Fall POSTECH Applied Linear Algebra) Let $V \subset \mathbb{R}^3$ be a two-dimensional subspace.

- (a) Show that the reflection of any vector $\mathbf{x} \in \mathbb{R}^3$ across V is

$$(2\mathbf{P} - \mathbf{I})\mathbf{x},$$

where $\mathbf{P}$ is the orthogonal projection onto V . *Hint.* Consider an orthonormal basis of V and extend it to an orthonormal basis of $\mathbb{R}^3$.

- (b) Let

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

be the standard basis of $\mathbb{R}^3$. Let V be spanned by

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Find the matrix representation of reflection across V with respect to the standard basis.

Problem 35. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & -1 \\ 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 & 1 \end{pmatrix}.$$

- (a) Find the matrices $\mathbf{P}$, $\mathbf{U}$, and $\mathbf{L}$ in a decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

(b) Solve $\mathbf{Ax} = \mathbf{b}$ for $\mathbf{x}$, where

$$\mathbf{b} = \begin{pmatrix} -2 \\ 3 \\ 5 \\ 4 \\ 1 \end{pmatrix}.$$

Problem 36. (2019 Fall POSTECH Applied Linear Algebra) Find a nonzero matrix $\mathbf{A}$ whose null space is spanned by

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

and whose left null space is spanned by

$$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Problem 37. (2019 Fall POSTECH Applied Linear Algebra) Fit a quadratic function

$$f(t) = c_0 + c_1t + c_2t^2$$

to the four data points

$$(-1, 8), \quad (0, 8), \quad (1, 4), \quad (2, 16).$$

Explain why your answer is the best fit.

Problem 38. (2019 Fall POSTECH Applied Linear Algebra) Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{QR},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 39. (2019 Fall POSTECH Applied Linear Algebra) Let P_4 be the vector space of real polynomials of degree at most 4 in the variable t . The map

$$T : P_4 \rightarrow P_4, \quad T(p(t)) = p(t+1),$$

is linear.

(a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$\{1, t+1, t^2+1, t^3+1, t^4+1\}.$$

(b) Find $\mathbf{A}^{-1}$.

Problem 40. (2019 Fall POSTECH Applied Linear Algebra) Fit a quadratic function

$$f(t) = c_0 + c_1 t + c_2 t^2$$

to the four data points

$$(-1, 8), \quad (0, 8), \quad (1, 4), \quad (2, 16).$$

Explain why your answer is the best fit.

Problem 41. (2019 Fall POSTECH Applied Linear Algebra) Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 42. (2019 Fall POSTECH Applied Linear Algebra) For a positive integer n , let S_n be the set of linear combinations of

$$\sin x, \dots, \sin nx, \cos x, \dots, \cos nx,$$

that is,

$$S_n = \left\{ \sum_{k=1}^n (a_k \sin kx + b_k \cos kx) \mid a_k, b_k \in \mathbb{R} \right\}.$$

Then the map

$$D : S_n \rightarrow S_n, \quad D(f) = \frac{df}{dx},$$

is linear.

(a) Find the matrix representation of D with respect to the ordered basis

$$\{\sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin nx, \cos nx\}.$$

(b) Find the determinant and trace of D .

Problem 43. (2019 Fall POSTECH Applied Linear Algebra) Let T be a linear operator on $\mathbb{R}^5$ represented by

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 3 \\ 1 & 2 & -4 & 0 & 3 \\ 1 & 0 & -1 & 1 & 1 \\ 1 & 2 & -3 & 0 & 2 \end{pmatrix}.$$

(a) Show that the characteristic polynomial has the form

$$p_{\mathbf{A}}(\lambda) = -(\lambda - 1)^3(\lambda - a)^2,$$

and find a .

(b) Find bases for the eigenspaces $E_1(\mathbf{A})$ and $E_a(\mathbf{A})$. Here, $E_1(\mathbf{A})$ (respectively, $E_a(\mathbf{A})$) denotes the subspace of $\mathbb{R}^5$ consisting of all eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 1 (respectively, a), together with the zero vector.

(c) Find the Jordan form of $\mathbf{A}$.

Problem 44. (2019 Fall POSTECH Applied Linear Algebra) Factorize

$$\mathbf{A} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1},$$

where $\mathbf{U}$ is unitary and $\mathbf{\Lambda}$ is diagonal.

Problem 45. (2019 Fall POSTECH Applied Linear Algebra) Suppose three cities A , B , and C are centers for trucks. Every month, half of the trucks in A and half of those in B go to C , while the other halves stay where they are. Every month, half of the trucks in C go to A and the other half go to B .

(a) Find the Markov transition matrix acting on the initial-state vector

$$\begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix},$$

where a_0, b_0, c_0 denote the initial numbers of trucks in cities A, B, C , respectively.

(b) If there are always 450 trucks, find the long-run distribution of trucks.

Problem 46. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}.$$

For what range of a is $\mathbf{A}$ positive definite?

Problem 47. (2019 Fall POSTECH Applied Linear Algebra) Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

Problem 48. (2019 Fall POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix.

(a) Show that $\mathbf{A} = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)^n$$

and $\mathbf{A}$ is Hermitian.

(b) Show that $\mathbf{A}^n = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)^n.$$

Hint. Consider the Jordan form.

Problem 49. (2019 Fall POSTECH Applied Linear Algebra) Let λ_0 be an eigenvalue of a square matrix $\mathbf{A}$, and let m be the algebraic multiplicity of λ_0 in the characteristic polynomial of $\mathbf{A}$. Show that

$$\dim E_{\lambda_0}(\mathbf{A}) \leq m.$$